

MATH 161A - Applied Probability & Statistics

Connor Petri

Fall 2025

Contents

1	Skipped	3
2	Probability	4
2.1	Sample Spaces and Events	4
2.1.1	Probability	4
2.2	Probability	7

1. Skipped

2. Probability

2.1 Sample Spaces and Events

Probability

- Probability is used as a mathematical tool to understand or describe random phenomena, chance variation, and uncertainty.
- We will use probability as a tool to evaluate the reliability of our conclusions about the population when we only have sample information.
- Probability is used in a variety of fields including genetics, quality control, and finance.
- Probability is used as a model for situation for which outcomes occur randomly.
- Such situations are called **experiments**.
- Examples:
 - Toss a coin.
 - Roll a die.
 - Draw a card from a deck.

Vocabulary

- An action or process whose outcome is subject to uncertainty is called an **experiment**.
- The possible outcomes of an experiment are called **sample space**.
- An **event** is a specific outcome or a set of outcomes from an experiment.
- A **simple event** is an event that consists of a single outcome from sample space S .

Example What is the outcome of rolling an even (E), rolling greater than 3 (F), and rolling a 2 (G) on a 6-sided die?

$$S = 1, 2, 3, 4, 5, 6$$

$$E = 2, 4, 6$$

$$F = 4, 5, 6$$

$$G = 2$$

A Note on Venn Diagrams All Venn Diagrams in this course require a rectangle to represent the sample space. Events are represented by circles within the rectangle.

Operations

Set Operations

- $A \subseteq B$: A is a subset of B .
- $A \cap B$: the intersection of A and B (elements in both A and B).
- $A \cup B$: the union of A and B (elements in either A or B).
- A^c or A' : the complement of A (elements in S that are not in A).

Logical Operations

- A or B : $A \vee B$
- A xor B : $A \oplus B$
- A and B : $A \wedge B$
- not A : $\neg A$

Mutually Exclusive Events

- Events A and B are **mutually exclusive** if they cannot both occur at the same time.
 - In terms of set operations: $A \cap B = \emptyset$.
 - Example: Rolling a die and getting an even number (E) or an odd number (O) are mutually exclusive events.
-

Identities

$$A \cup A' = S$$

$$A \cap A' = \emptyset$$

DeMorgan's Laws

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

$$\neg(A \vee B) = \neg A \wedge \neg B$$

$$\neg(A \wedge B) = \neg A \vee \neg B$$

2.2 Probability

Definition Probability is a function that maps the sample space S to $\mathbb{R}$. The sample space is the **domain** and the real numbers are the **codomain**.

Axioms of Probability

1. Non-negativity: For any event A , $P(A) \geq 0$.
2. Normalization: $P(S) = 1$, where S is the sample space.
3. Additivity: For any two mutually exclusive events A and B , $P(A_1 \cup A_2 \cup \dots \cup A_n) = \sum_{i=1}^n P(A_i)$.

$$P(\cup_{i=1}^n A_i) = \sum_{i=1}^n P(A_i)$$

Not axioms but useful observations

1. $P(A) < 1$ for any event A .
2. $P(A^c) = 1 - P(A)$.
3. $P(A \cap B) = P(A) + P(B) - P(A \cup B)$ for any two events A and B .

Proving Observations

Observation 1: $P(\emptyset) = 0$

Proof. $P(\emptyset) = 0$

$$\emptyset \cup S = S$$

$$\emptyset \cap S = \emptyset$$

$$P(\emptyset \cup S) = P(S)$$

$\emptyset$ and S are mutually exclusive.

$$P(\emptyset) + P(S) = P(S) \text{ by axiom 3.}$$

$\therefore P(\emptyset) = 0$ by algebra. □

Observation 2: $P(A^c) = 1 - P(A)$

Proof. $S = A \cup A^c$

$$\emptyset = A \cap A^c$$

$$P(A^c \cup A) = P(S)$$

A and A^c are mutually exclusive by line 2.

$$P(A^c) + P(A) = P(S) \text{ by axiom 3.}$$

$$P(A^c) + P(A) = 1$$

$$\therefore P(A^c) = 1 - P(A) \text{ by algebra.}$$

□

Observation 3: $P(A \cap B) = P(A) + P(B) \implies A \cap B = \emptyset$

Proof. $P(\cup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} P(A_i)$ for any countable collection of mutually exclusive events A_i .

$$\text{Let } A_1 = A, A_2 = B, A_3 = A_4 = \emptyset. P(A \cup B \cup \emptyset \cup \emptyset) = P(A) + P(B) + 0$$

$$\therefore P(A \cup B) = P(A) + P(B) \text{ by axiom 3.}$$

□

Identities for the next proof

1. $P(A) = P(A \cap B') + P(A \cap B)$
2. $P(B) = P(A' \cap B) + P(A \cap B)$
3. $P(A \cup B) = P(A \cap B') + P(A \cap B) + P(A' \cap B)$
4. $1 = P(A \cap B') + P(A \cap B) + P(A' \cap B) + P(A' \cap B')$

Observation 4: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Proof. $P(A \cup B) = P(A \cap B') + P(A \cap B) + P(A' \cap B)$ by 3.

$$P(A \cup B) = P(A) + P(A' \cap B) \text{ by 1.}$$

$$P(A \cup B) = P(A) + [P(B) - P(A \cap B)]$$

$$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B) \text{ by algebra.}$$

□

Other Propositions

$$(A \subseteq B \implies P(A) \leq P(B))$$

Proof. $B = (A \cap B) \cup (A' \cap B)$

$$A \cap B = A$$

$$B = A \cup (A' \cap B)$$

$$P(B) = P(A) + P(A' \cap B) \text{ by axiom 3.}$$

$$P(B) \geq P(A) \text{ since } P(A' \cap B) \geq 0 \text{ by axiom 1.}$$

For any A , $A \subseteq S$.

$$P(A) \leq P(S).$$

$$\therefore P(A) < 1.$$

□